

An Effective Fault Detection Technique for Grid-Connected Electric Vehicle Charging Station

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Abstract—The widespread integration of electric vehicle (EV) charging stations (CS) into the existing utility grid is giving rise to numerous power quality and protection challenges, significantly impacting load performance at both large and small consumer ends. This phenomenon is escalating as the transition from internal combustion engine (IC) vehicles to EVs gains momentum, driven by their inherent advantages. EV chargers utilize power electronics to convert AC power into DC power for battery charging from the grid. However, this conversion process, facilitated by high-frequency switching converters, introduces harmonics into the grid, consequently deteriorating the power quality of grid line currents. These injected harmonics lead to electrical and thermal overloading of distribution transformers, thereby compromising their lifespan. Additional issues include excessive current draw from the grid, imbalance between demand and supply, and voltage fluctuations. Apart from power quality issues, there is a need to develop a fast fault detection algorithm to protect the charging station supply equipment when a fault occurs either grid side or on the downstream DC side. This paper presents an instantaneous power-based fault detection algorithm to detect the fault in grid-connected EV charging stations for Grid vehicle (G2V) and vehicle-to-grid (V2G) mode of operation.

Keywords—Abs (Absolute value), G2V (grid to vehicle), EV (Electric Vehicle), Pint (Instantaneous power), Q-sum (Cumulative sum), V2G (vehicle to grid)

I. INTRODUCTION

THE presence of electric vehicles (EVs) has become an unavoidable component of future power systems. The extensive incorporation of EV charging stations (EVCS) into the distribution grid has repercussions on power quality, including impacts on power demand, voltage imbalances, system stability, harmonic pollution, and other adverse effects. [1], [2]. Significant hurdles involve safeguarding the life cycle of EV batteries and protecting EVCS supply equipment amidst various types of voltage sags resulting from transient disturbances in three-phase AC power lines [3]. In the literature, various initiatives have been suggested to mitigate the adverse effects of EVs on the utility grid. Research cited in Reference [4] indicates that even a low penetration of EVs can be problematic if not managed in a controlled manner. The impact of a PHEV (plug-in hybrid electric vehicle) on a low-voltage distribution transformer is adverse when it is overloaded. Furthermore, it also affects transformer aging and distorts the voltage supply [5]. Recent enhancements in AC and DC fast-charging standards have been notably advanced by IEEE, IEC, and SAE bodies [6]. With the expansion of high-charging equipment accessibility, public fast charging stations, and the advancement of renewable energy sources (RES)-based EV charging, we anticipate exponential growth in EV sales. Efforts are primarily directed toward enhancing RES power charging infrastructures to leverage EV benefits while bolstering grid operations [7]. The chopper D.C to D.C converter design is crucial for EV charging and discharging in

charging infrastructure. The design of a fuzzy logic controlled-based shunt converter for the charging scheme of EVs is illustrated in [8]. The stability of EV chargers is influenced by modes of operation and grid parameters, which also explains about critical voltage to keep the charger in stable mode [9]. [10] describes the impact of grid disturbances on battery chargers during G2V, V2G, and V2H operations. A voltage sag compensation scheme is demonstrated in [11], which also discusses a pre-sag compensation strategy that precisely maintains the sensitive load voltage at its pre-fault value during a common bus voltage sag.

The main contribution of the author for this article is as follows:

- To develop an EV charging station that will work for both G2V and V2G modes.
- Development of an effective fault detection technique to safeguard the grid-connected EVCS from upstream A.C side fault.
- To enhance the adaptiveness of the algorithm for high-impedance faults.
- The algorithm can detect multiple A.C faults occurring on the upstream side of EVCS with minimum time.

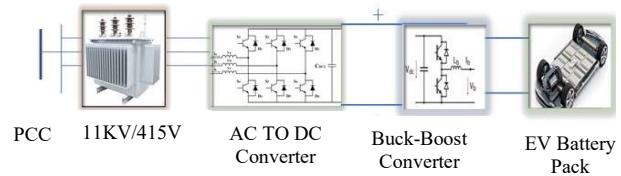


Fig. 1. Schematic of the proposed grid-connected EVCS.

II. GRID CONNECTED EVCS

In this article, the author has proposed an electric vehicle charging station mainly consisting of a bidirectional AC to DC or DC to AC-controlled converter, buck-boost bidirectional converter, and EV battery to be charged as shown in the Fig.1. The interfacing inductor is used to connect the converter with the low voltage distribution 415V grid to smooth the current ripple present in the input side grid current. The use of the DC link capacitor regulates the DC output voltage of ac to DC converter. Further DC voltage is stepped down using a buck converter to charge the EV battery at the desired voltage. In this system, two controllers such as the converter controller and the chopper controller are used. The concept of synchronous reference frame (SRF) theory is used to design the converter controller to draw the AC power from the grid at unity power factor, and also maintain the DC link output voltage at the desired value. The chopper controller is used to generate the gate pulses for the IGBT

switches to step down the voltage in the G2V mode of operation and step up the DC voltage during the V2G operation. The mathematical formula to calculate the DC output voltage is following:

$$V_{dc} = \frac{2\sqrt{2}V_{L-L}}{\sqrt{3}m} \quad (1)$$

Where V_{L-L} is line-to-line voltage in volts, m is the modulation index ranges between 0.8 to 0.9. The design of the DC link capacitor filter to get regulated DC output voltage is done [3]. and the proper value of the DC link capacitor is selected using the following formula

$$C = \frac{P_{dc}}{V_{dc} * 4\pi * f * (V_2 - V_1)} \quad (2)$$

The chopper capacitor and inductor values for the buck-boost converter are calculated using [3], [8].

$$L_b = \frac{D(V_{dc} - V_b)}{f_s * \Delta I_L} \quad (3)$$

$$C_b = \frac{\Delta I_L}{8 * f_s * \Delta V_b} \quad (4)$$

The rectifier control strategy is done using SRF theory which utilizes V_{abc} , I_{abc} , and V_{dc} as input. By use of abc to dqo converter, PLL, PI controller and PWM pulse generator, a gate pulse has been generated for the rectifier converter to get the desired output voltage. Further chopper control technique is used to control the charging current of the EVCS.

TABLE-I
SYSTEM PARAMETERS FOR 14 kW EVCS

Sl. No.	Parameter	Symbol	Value
1	Supply bus voltage	V_s	415 V (L-L)
2	Power grid frequency	f	50 Hz
3	Output Filter inductor	L_f	5 mH
4	Output Filter capacitor	C_f	3300 μ F
5	DC link capacitor voltage	V_{dc}	750 V
6	Buck-boost Capacitor	C_b	50 μ F
7	Buck-boost inductor	L_b	4 mH
8	EV Battery nominal voltage	V_o	360 V
9	EV Battery capacity	Q	66.2 Ah
10	Converter Switching frequency	f_s	10 kHz

The system parameters for the designed EVCS are shown in Table I.

III. PROPOSED TECHNIQUE

The three-phase voltage and current signals are collected from PT and CT sensors respectively. Using phasor estimation voltage and Current rms phasor are calculated. In this paper, the author has proposed a fault detection technique using instantaneous power for both AC and DC sides of grid-

connected EVCS. The instantaneous power is calculated using discrete voltage and current signals, and it is given by equation (5).

$$p(n) = v_a(n) * i_a(n) + v_b(n) * i_b(n) + v_c(n) * i_c(n) \quad (5)$$

The cumulative sum of the absolute value of three-phase instantaneous power is calculated for post-fault samples. The cumsum value for post fault is now compared with the selected threshold value and the trip decision is taken as fast as possible. The cumsum-based instantaneous power fault detection technique algorithm flow chart is shown in Fig. 2. The value of the threshold is tested for high impedance fault too. It is found that the algorithm takes more time to detect the high impedance fault that occurs on the AC side of the charging station. To enhance the adaptiveness of the detection algorithm, the selected threshold value is tested for sensor data which is mixed with Gaussian noise of 30db. The noisy sample data is taken to calculate the power and hence cumsum of absolute value of instantaneous power is then calculated. Whenever any type of fault occurs on the upstream side of the charging station the value of instantaneous power will be increased enormously. The cumsum value is calculated for the absolute value of this increased instantaneous power. Before taking the trip decision we need to take care of transients, EV switching, or any load switching to avoid nuisance tripping using a timer circuit.

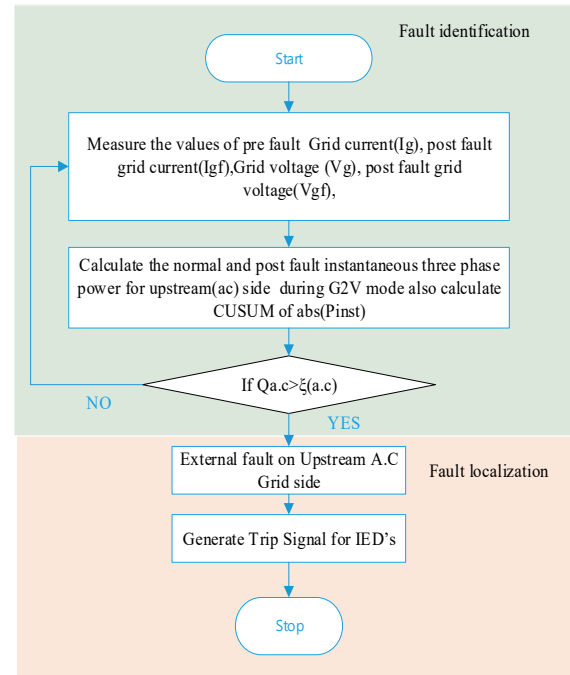


Fig. 2. Flow chart for fault detection technique.

If the cumulative sum of instantaneous power is greater than the threshold value then a fault has been detected. The location of the fault in EVCS is explained using Table II. The three-phase grid voltage and current during different types of AC fault conditions of a grid-connected electric vehicle charging station are shown in section IV. The three-phase current sensor data is stored during normal conditions for the G2V mode of operation. The charging behavior and performance characteristics are different during a fault occurs in the grid

and it can deteriorate the battery life cycle of the EV. It also leads to safety issues for electric vehicle supply equipment connected to EVCS. The value of the threshold is selected 2.57×10^8 to detect the fault as quickly as possible.

IV. RESULTS AND DISCUSSION

A. Three phase to Ground fault (LLLG fault)

A three-phase to-ground fault with zero fault impedance is created in the three-phase AC upstream side. The discrete sample sensor data is stored to verify the proposed algorithm and a trip signal has been generated for this scenario. The three phases to ground fault have been simulated between 0.2 sec to 0.7 sec shown in Fig. 3. In this case the three-phase voltage suddenly drops to zero at 0.2 sec due to this grid current has increased enormously high and hence instantaneous power drawn has increased. The DC link capacitor voltage will drop due to three phases and also high ripple prevails during fault. The charging behavior will be affected due to a drop in the DC link

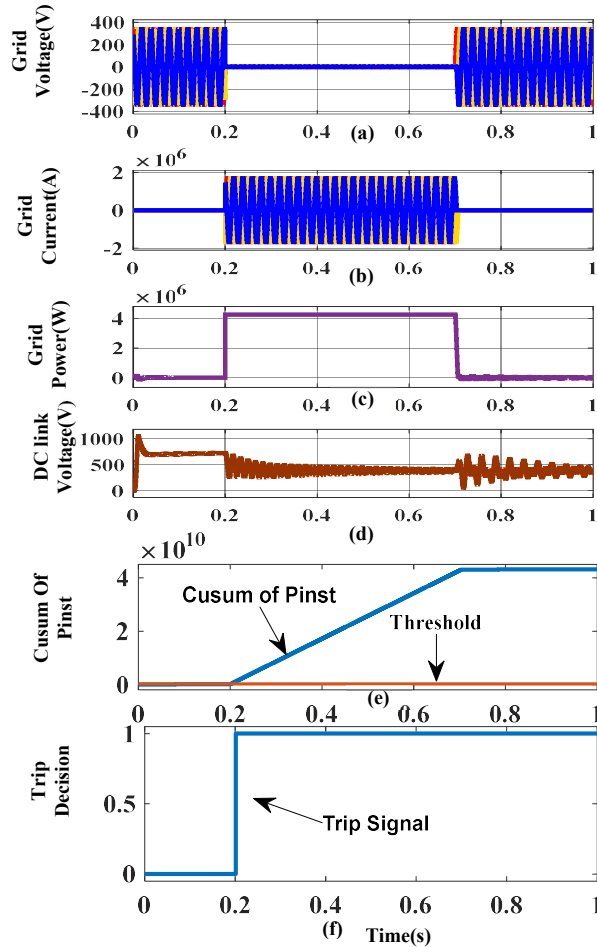


Fig. 3. EVCS characteristics during LLLG fault with $R_f=0 \Omega$ (a) Grid voltage (b) Instantaneous power consumed (c) Grid current (d) DC link output voltage (e) Cumsum of Pinst and threshold (f) Trip signal.

Voltage is shown in Fig. 3. The sum of the absolute value of instantaneous power is calculated and compared with a

threshold value, and a trip signal is generated as depicted in Fig. 4.

B. Line to Ground fault (LG fault)

Line to ground fault is initiated at 0.2 sec and cleared at 0.7 sec. The charging behavior as well as grid characteristics are observed and data samples are collected at the sampling rate of 50e-6 samples/sec. The three-phase grid current, voltage, instantaneous power, and DC link capacitor voltage are shown in Fig. 4. The instantaneous power and three-phase grid current are less as compared to three three-phase short circuit faults. The pictorial details found through MATLAB simulation of above above-discussed LG fault are depicted in Fig. 4.

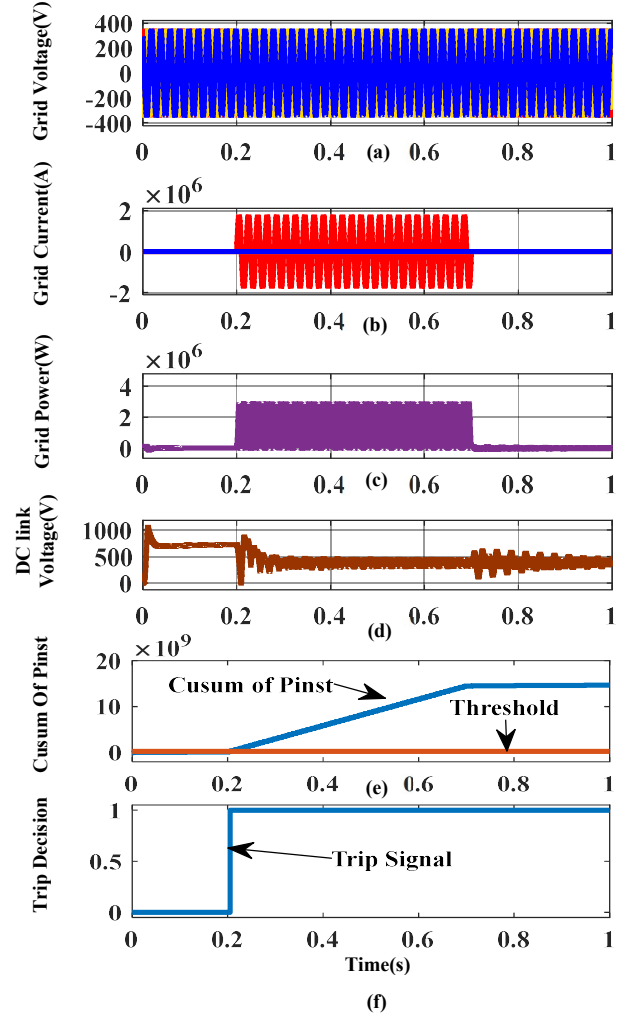


Fig. 4. Grid characteristics during LG fault with $R_f=0 \Omega$ (a) Grid voltage (b) Instantaneous power consumed (c) Grid current (d) DC link output voltage (e) Cumsum of Pinst and threshold (f) Trip signal.

The ripple in the DC link capacitor is sustained more as compared to a three-phase dead short circuit fault. The cumsum of instantaneous power is calculated and compared with the selected threshold. It found that during fault cumsum of power is greater than the threshold and hence trip signal is generated by concern IED. The cumsum during the LG fault and the trip

signal are shown in Fig. 6. The algorithm takes 5.05ms to detect the LG fault occurs in the three-phase 415V upstream side. When fault impedance increases the fault detection time taken by IEDs will be more. It is found that the occurrence of LG faults is most frequent as compared to other faults, so it is necessary to have an adaptive fault detection algorithm during LG faults.

C. Line to Line fault (LL fault)

The grid performance of EVCS during an upstream AC side line-to-line fault is shown in Fig.5. The grid current, instantaneous power, and DC link capacitor voltage have increased enormously high when a line-to-line fault occurs on the AC side. The fault initiated at 0.2 sec and cleared at 0.7 sec and the duration of the fault is 0.5sec. The ripple in DC link voltage is increased during the fault.

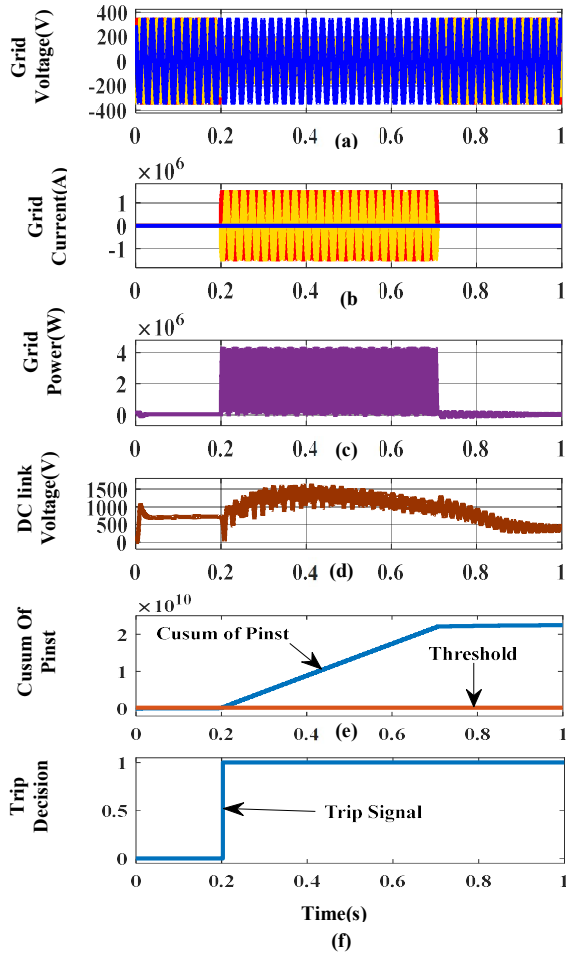


Fig. 5. Grid characteristics during LL fault with $R_f=0 \Omega$ (a) Grid voltage (b) Instantaneous power consumed (c) Grid current (d) DC link output voltage (e) Cumsum of Pinst and threshold (f) Trip signal.

Finally, the trip signal is generated as per the fault detection algorithm based on the cumsum value of instantaneous power depicted in Fig. 5. The fault detection time for line-to-line fault is observed as 1.4 ms when fault impedance is zero.

D. Double Line to Ground fault (LLG fault)

The double line to ground fault is simulated between 0.2 sec to 0.7 sec with a total simulation time of 1 sec in

MATLAB/SIMULINK software. During LLG fault, the instantaneous power and current have been increased while the reduction in DC link capacitor voltage is observed. The above-discussed characteristics are depicted via Fig.9. The LLG fault affects the EV charging behavior due to a reduction in DC link voltage.

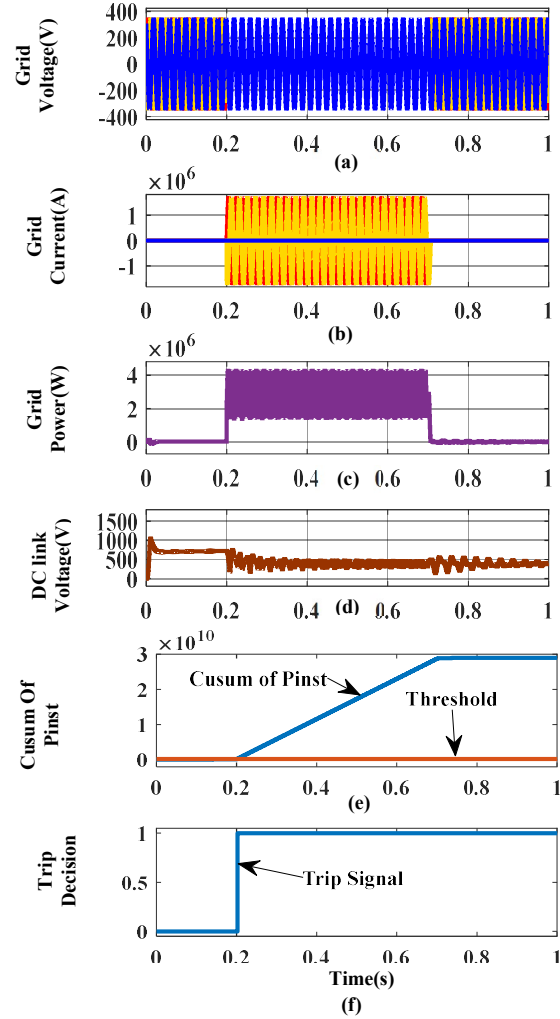


Fig. 6. Grid characteristics during LLG fault with $R_f=0 \Omega$ (a) Grid voltage (b) Instantaneous power consumed (c) Grid current (d) DC link output voltage (e) Cumsum of Pinst and threshold (f) Trip signal.

So, there is a need to detect the fault as quickly as possible. The proposed algorithm calculates the cumsum of instantaneous power and compares it with a threshold value. If the condition for fault criteria meets then a trip signal has been generated.

TABLE-II
FAULT LOCALIZATION

SL. No.	Condition	Fault detection	Fault location
1	$Q_{ac} < \xi_{ac}$	No	-
2	$Q_{ac} > \xi_{ac}$	yes	AC Grid side

Fig. 6. of (e) and (f) depict the whole trip signal generation process. The LLG fault is detected in 0.85ms as per the proposed algorithm. The comparison of fault detection time taken by the proposed algorithm for different types of AC faults occurs in the grid-connected charging station explained in table-III.

TABLE-III
FAULT DETECTION TIME FOR DIFFERENT FAULTS

SL NO.	Parameters	LG	LLG	LL	LLG
1	Maximum Q-sum value	1.4642e+10	4.3206e+10	2.2441e+10	2.9035e+10
2	Maximum power consumed (MW)	2.891449	4.263485	4.264765	4.263485
3	Fault detection time(ms)	5.05	0.1	1.4	0.85

V. CONCLUSION

This article introduces an effective technique for detecting the AC side faults in grid-connected electric vehicle charging stations. The author has also focused on developing EVCS which can work for both G2V and V2G modes of operation. The fault detection technique can identify the different types of faults that occur on the AC side. The algorithm's robustness is tested for low to high-impedance faults occurring on the upstream side. The fault detection time for different types of AC faults is compared in table-III. It shows that the algorithm detects the LLLG fault in 0.1ms, which is very fast compared to other faults. A trip signal is generated based on the cumsum of instantaneous power for different faults. The severity of the LLLG fault is the most severe and the LG fault is the least among all AC faults occurring on the upstream side of the charging station. The algorithm takes more detection time in case of LG fault.

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